

A Two-Level Grammar of Nominal Potential

Lexical Structure, Syntactic Slots, and the Determiner Status of NOT

Expanded edition

Monograph

January 08, 2026

Author: Maks Panner

Affiliation: Independent Researcher in Linguistic Theory

Abstract

This monograph develops a two-level architecture for grammatical description that separates (i) a lexical module responsible for head category, phrase-internal selection, and local constituency, from (ii) a syntactic module responsible for sentence roles (Subject–Predicate–Object) and the slot families that license constituents into those roles. The empirical focus is English NOT.

While not is often treated as a clausal negator in the functional spine, it also appears in a broad family of constituent uses with complements that are heterogeneous in lexical category (NPs, infinitivals, finite clauses in argumental packaging, PPs, APs, and participial/eventive phrases).

The central generalization proposed here is that this complement diversity is not category-driven but licensing-driven: NOT targets exactly those constituents that are independently admissible as nominal candidates in the syntactic role system. I formalize this by splitting NP into NP_lex (noun-headed phrases computed in the lexical module) and NP_syn (the functional class of constituents with nominal potential in the syntactic module). Constituent NOT is analyzed as a syntactic determiner-like operator Det_syn that selects NP_syn and returns NP_syn (NP_syn → NOT NP_syn). This analysis predicts (a) non-competition with articles and other Det_lex items, (b) recursion under NP_syn, (c) a sharp separation between constituent-NOT distribution and clause-level negation diagnostics (auxiliary support, inversion, polarity licensing), and (d) typological coupling between NP_syn expansion (nominalization resources) and the availability of constituent-negation operators. The result is a compact formal grammar with explicit interface principles and an empirically testable program for diagnosing nominal potential.

Keywords

two-level grammar; nominal potential; NP split; syntactic determiner; constituent negation; clausal negation; polarity licensing; typology; formal grammar; interface

Table of Contents

1. Preface and Scope
2. Introduction
3. Background: Categories, Functions, and Negation
4. The Lexical Module
5. The Syntactic Module: Roles and Slot Families
6. Diagnostics for Nominal Potential and the NP Split
7. NOT as a Syntactic Determiner (Det_syn)
8. Distribution, Constraints, and Markedness
9. Formal Grammar and the Lexical–Syntactic Interface (BNF)
10. Worked Derivations
11. Interpretation: Nominal Potential as Type Access
12. The Semantics of NP_syn-level NOT
13. Cross-Linguistic Calibration
14. Why Uniform NP/DP Implementations Overgenerate
15. Diachrony: Negation and Nominalization
16. Computational and Psycholinguistic Implications
17. Conclusion and Future Work
18. Construction Atlas: A Glossed Example Set by Construction Type
19. Constituent NOT vs Clausal Negation: Auxiliaries, Scope, and Polarity
20. Typology: NP_syn Diagnostics Applied Language-by-Language
21. Related Work and Distinguishing Predictions
22. Computational Applications
23. Neuroscience and Neurocognitive Implications

Appendix A. Notation

Appendix B. Master BNF Specification

Appendix C. Extended Derivations

Appendix D. Glossing Abbreviations

References

1. Preface and Scope

This book is a methodological argument with an empirical anchor. The methodological claim is that grammatical theory benefits from a disciplined separation between two kinds of description that are routinely conflated: category and function. Category is computed in a lexical module: it determines what a head is, what it selects, and how phrase-internal structure is assembled.

Function is computed in a syntactic module: it determines which constituents occupy subject, predicate, and object roles (and their discourse refinements), and it provides the slot families that license constituents into those roles.

The empirical anchor is the English item NOT. Its distribution is unusually revealing because it combines not only with canonical noun phrases (not a cat) but also with infinitivals (not to swim), subordinate clauses used as arguments (not that he left), PPs in contrastive/topic positions (not in the house), APs (not ready), and participial phrases (not sitting in silence). These complements share neither a head category nor a uniform internal morphology. What they share is that they can be hosted as nominal candidates - units that a clause can place into subject/object/topic slots and treat as discourse objects.

Cross-linguistic remarks are used as calibration and as a source of predictions, not as exhaustive typology. Throughout, descriptive claims (what the system must represent to capture the data) are separated from implementation choices (how a particular formalism encodes those claims).

2. Introduction

2.1 Category conflation and the NP problem

Many frameworks use NP ambiguously: sometimes as a noun-headed phrase (a lexical category), and sometimes as whatever fills the subject/object position (a syntactic function). This ambiguity becomes visible the moment we observe that noun-headed phrases can occur as predicates (He is a doctor), and that non-noun constituents can occupy subject slots (To leave now would be a mistake).

If NP is treated as a single object, nominalization looks like a mysterious operation that turns non-nominals into nominals. This pushes analyses toward silent structure: covert N heads, covert D heads, or nominalizing projections inserted whenever a clause-like or phrase-like constituent behaves as an argument. The present approach avoids both the mystery and the covert proliferation by treating nominal behavior as a property of syntactic hosting rather than lexical headedness.

2.2 The empirical trigger: NOT

English not is commonly introduced as a clausal negator. Yet its constituent uses are broad. Consider the following patterns, where the bracketed constituents are not homogeneous in lexical category:

- (1) a. not [a cat]
b. not [to swim]
c. not [that he left]
d. not [in the house]
e. not [ready]
f. not [sitting in silence]

The goal is to state a single generalization that covers (1a-f) without multiplying homophonous nots, and without forcing covert nominal structure.

2.3 Contributions

The principal contributions are: (i) a two-level architecture separating lexical selection from syntactic role licensing; (ii) a definition of nominal potential as a syntactic property independent of noun-headedness; (iii) a formal split between NP_lex (noun-headed phrases) and NP_syn (nominal-slot candidates); (iv) a determiner-level analysis of NOT as Det_syn, capturing its distribution and its non-competition with articles; and (v) a compact formal grammar (BNF) plus derivations and an interpretation component sufficient to state scope, recursion, and selection facts with explicit interface principles.

3. Background: Categories, Functions, and Negation

3.1 Lexical categories and selectional profiles

Lexical categories (N, V, A, P, etc.) are defined by distribution, morphology, and selection. Heads select complements as part of a valency profile (Baker 2003). Selection is phrase-internal; it does not presuppose sentence roles. A verb may select an NP_lex, a PP_lex, or a clausal complement; whether that complement ultimately appears as an object, a subject, or a topic is a matter of syntactic hosting.

3.2 Syntactic functions and role schemas

Subject, predicate, and object are syntactic functions (roles). Languages encode these roles via word order, agreement, case, and information-structural marking. Methodologically, functional labels should not be used as if they were lexical categories. Treating whatever is in subject position as an NP is harmless only if NP is explicitly a functional class - precisely the role played here by NP_syn.

3.3 The NP/DP tradition and its limits

The DP hypothesis treats articles as heads of a functional projection DP that takes NP as complement (Abney 1987). This is useful for describing article syntax in noun-headed phrases. However, DP theory does not by itself solve the nominalization problem: it does not explain why non-noun constituents behave nominally in subject/object positions. Implementations often respond by adding silent nominal structure (covert D, covert N, nominalizing heads). The risk is

descriptive non-falsifiability: if any constituent can be made a DP by hypothesis, the analysis ceases to predict where nominal behavior should occur.

3.4 Negation families

Negation is often divided into clausal negation, constituent negation, and polarity licensing (Horn 1989; Haegeman 1995; Zeijlstra 2004). English not participates in all three. The present monograph isolates an NP_syn-targeting use of not and shows that this use provides a diagnostic for nominal potential as a licensing condition separate from clausehood.

4. The Lexical Module

4.1 Inventory and phrase-internal structure

The lexical module supplies (i) a set of lexical categories, (ii) head-driven phrase structure rules, and (iii) selectional constraints. A minimal inventory sufficient for the present purpose is N, V, A, P, and Det_lex.

4.2 NP_lex: noun-headed phrases

Definition 1 (NP_lex). NP_lex is a phrase headed by N. Articles and possessives are Det_lex elements that combine with nouns in the lexical module. NP_lex may contain adjectival modifiers and PP modifiers, subject to standard constraints.

```
Lexical schemata (informal):  
NP_lex  -> (Det_lex) N (Mod_lex) *  
Mod_lex -> AP_lex | PP_lex  
PP_lex  -> P NP_lex  
AP_lex  -> A (PP_lex) ?
```

4.3 Lexical complements and internal argument structure

Complementhood is a lexical notion: a complement is selected by a head as part of its valency profile. This matters because it prevents a common slide: treating object as if it were complement. A verb can select a clause complement; that clause may later function as an object, a subject, or a topic in syntax. The lexical module cannot decide that; it only constrains selection possibilities.

4.4 Why NOT is not lexical

NOT does not behave like Det_lex. It is not restricted to nouns; it does not occupy the article slot inside NP_lex; it does not participate in determiner systems as a member of the article paradigm; and it can attach to non-noun complements. Treating NOT as lexical therefore forces either multiple homophonous NOTs or extensive covert structure. A syntactic-operator analysis avoids both.

5. The Syntactic Module: Roles and Slot Families

5.1 Role primitives

The syntactic module is defined by sentence roles. For present purposes, the core schema is Subject-Predicate-Object (SPO), where Predicate is the predication center and Subject/Object are argumental slots licensed by it. Refinements (topic, focus, oblique roles) are treated as structured sub-slots or alternative hosting sites, but remain governed by the same licensing logic.

5.2 Slot families and admissible hosts

Each role induces a family of slots. The crucial claim is that slot families are typed by function, not by lexical category. Nominal slots do not require noun headedness; they require nominal potential. This is why infinitivals, clauses, PPs, APs, and participial phrases can occupy nominal slots in English under appropriate constructions.

5.3 Nominal potential

Definition 2 (Nominal potential). A phrase has nominal potential iff it is licensed to occupy a nominal slot in the syntactic module (subject, object, or a discourse-nominal position such as topic). Nominal potential is a property of interface licensing. It can be diagnosed by (i) distribution in nominal slots, (ii) compatibility with nominal proforms and anaphora in relevant constructions, and (iii) compatibility with NP_syn-level operators such as constituent NOT.

5.4 NP_syn: the functional nominal class

Definition 3 (NP_syn). NP_syn is the class of phrases with nominal potential. NP_syn is not a single lexical category; it is a family of admissible host types. NP_lex is included as one member, but NP_syn also includes infinitivals, certain clauses, PPs, APs, and participial phrases when used argumentally or topically.

6. Diagnostics for Nominal Potential and the NP Split

6.1 Hosting diagnostics

The most direct test for nominal potential is hosting: can the phrase occupy a subject slot, object slot, or topic slot without overt nominal morphology?

- | | |
|--|---------------------------|
| (2) a. [To swim] is fun. | (Infinitival subject) |
| b. I regret [to leave so early]. | (Infinitival complement; |
| register-sensitive) | |
| c. I remember [that he left]. | (Clausal complement) |
| d. [In the garden] is where I found it. | (PP in focus/topic frame) |
| e. [Ready] is how he seemed. | (AP in nominal slot via |
| wh-cleft) | |
| f. [Sitting in silence] was the problem. | (Participial/eventive |
| nominal) | |

6.2 Operator diagnostics

A second diagnostic is the availability of NP_syn-level operators: operators that do not care about noun-headedness but care about nominal hosting. Constituent NOT is precisely such an operator. If NOT can precede a phrase and the result can occupy a nominal slot, that phrase is a strong candidate for NP_syn membership.

6.3 Identity and re-identifiability

Because nominal potential is a licensing property, it must be trackable under discourse operations (topicalization, coordination, ellipsis). NP_syn is therefore treated as a re-identifiable host class: once a phrase is admitted as NP_syn in a derivation, it remains eligible for nominal tracking.

6.4 The NP split

The split is thus: NP_lex is defined by headedness and computed in the lexical module; NP_syn is defined by hosting eligibility and computed in the syntactic module.

7. NOT as a Syntactic Determiner (Det_syn)

7.1 The core rule

The central generalization can be stated as a single licensing rule in the syntactic module:

```
Det_syn ::= NOT
NP_syn  -> Det_syn NP_syn
    i.e., NP_syn -> NOT NP_syn
```

NOT selects NP_syn (nominal potential), not NP_lex (noun headedness). The output is again NP_syn.

7.2 Non-competition with articles

Because NOT is Det_syn rather than Det_lex, it does not compete with a/the/my. This predicts the basic asymmetry:

- (3) a. not [a cat]
- b. *a [not cat]
- c. not [the cat]
- d. *the [not cat]

7.3 Why NOT is determiner-like

Determiners are operators that package a nominal candidate for hosting and quantificational/selectional behavior. NOT, in the constituent use at issue, packages a nominal candidate as excluded relative to a discourse contrast set. It is therefore determiner-like at the syntactic level.

8. Distribution, Constraints, and Markedness

8.1 Distribution over NP_syn members

NOT combines with NP_lex and with non-noun NP_syn members in the constructions that license them as nominal candidates.

- (4) a. not [a cat] / not [my neighbor]
b. not [to swim]
c. not [that he left] / not [because he was tired]
d. not [in the house] / not [on Tuesday]
e. not [ready] / not [happy]
f. not [sitting in silence]

8.2 Constraint: NOT is external to NP_lex

The theory predicts that NOT cannot appear inside NP_lex as a lexical determiner competitor. NP_syn-level NOT can take NP_lex as its complement, yielding not [Det_lex N], but the reverse configuration is excluded.

8.3 Construction sensitivity and pragmatic markedness

Where NP_syn membership is construction-dependent, NOT's acceptability tracks that dependence. The claim is testable: when a phrase type is licensed as a nominal candidate, NOT is available in principle; when a phrase type is not licensed as a nominal candidate, NOT is blocked.

8.4 Iteration and recursion

NP_syn recursion permits nested NOT in principle (not (not X)). Iteration is often pragmatically marked, but its structural availability argues for an NP_syn-level operator.

9. Formal Grammar and the Lexical–Syntactic Interface (BNF)

9.1 Lexical grammar

Lexical categories:

N, V, A, P, Det_lex

Det_lex ::= a | the | this | that | these | those | my | your | his | her
| its | our | their | Ø

NP_lex ::= (Det_lex) N (Mod_lex)*

Mod_lex ::= AP_lex | PP_lex

PP_lex ::= P NP_lex

AP_lex ::= A (PP_lex)?

VP_lex ::= V (NP_lex)? (PP_lex)* (Clause_lex)?

9.2 Syntactic grammar

Role schema:

```
S ::= Subject Predicate (Object)
```

```
Subject    ::= NP_syn
```

```
Object     ::= NP_syn
```

```
Predicate  ::= VP_syn
```

```
Det_syn    ::= NOT
```

```
NP_syn     ::= NP_lex  
              | InfP_syn  
              | Clause_syn  
              | PP_lex  
              | AP_lex  
              | PartP_syn  
              | Det_syn NP_syn
```

```
InfP_syn   ::= to V (NP_syn)? (PP_lex)*
```

```
PartP_syn  ::= V-ing (NP_syn)? (PP_lex)*
```

```
Clause_syn ::= that S | because S | when S | if S
```

```
VP_syn     ::= V  
              | V NP_syn  
              | V PP_lex  
              | V NP_syn PP_lex  
              | V Clause_syn
```

9.3 Interface principles

The interface imports lexical phrases into syntactic hosting positions. NP_lex is always admissible as NP_syn. Other phrase types become NP_syn only when a syntactic construction licenses them as nominal candidates. NOT is defined only in the syntactic module, ensuring its distribution is sensitive to role licensing rather than lexical selection.

10. Worked Derivations

10.1 To swim is fun

```
Subject    -> NP_syn -> InfP_syn -> to swim
```

```
Predicate  -> VP_syn -> is fun
```

10.2 Not to swim is dangerous

```
Subject    -> NP_syn -> NOT NP_syn -> NOT InfP_syn -> not to swim
```

```
Predicate  -> VP_syn -> is dangerous
```

10.3 Not that he left bothered me

Subject -> NP_syn -> NOT Clause_syn -> not that S
Predicate -> VP_syn -> bothered NP_syn
Object -> NP_syn -> me

10.4 Not in the house is where I found it

Topic/Subject -> NP_syn -> NOT PP_lex -> not in the house
Predicate -> is where I found it

10.5 Recursion

NP_syn -> NOT NP_syn -> NOT (NOT NP_syn) -> not (not X)

11. Interpretation: Nominal Potential as Type Access

11.1 Generalized argument type

Nominal potential can be modeled as access to a generalized argument type e^* - an argumental domain that can host individuals, events, propositions, locations, properties, and other discourse-manageable objects. Under this approach, NP_syn denotes elements of e^* ; NP_lex denotes individuals (or generalized quantifiers, depending on the semantic architecture). Non-noun NP_syn members are coerced into e^* when hosted in nominal slots.

11.2 Nominalization as licensing, not covert N

On this view, nominalization is not primarily a derivational operation that inserts a covert noun head. It is a set of licensing strategies that make a non-noun constituent eligible for e^* hosting.

12. The Semantics of NP_syn-level NOT

12.1 NOT as exclusion over e^*

In constituent use, NOT can be modeled as an exclusion operator over e^* : it maps a candidate x in e^* to an excluded variant relative to a contextually supplied contrast set. When x is an individual, the effect is individual exclusion; when x is an event or proposition, the effect is event/proposition exclusion.

12.2 Scope

Because NP_syn-level NOT is inside NP_syn, its scope is the nominal candidate itself. This distinguishes NP_syn-level NOT from clausal negation, which targets propositional content in the clausal spine.

12.3 Iteration

Iteration (not (not X)) is structurally available via NP_syn recursion and pragmatically constrained by contrast-set accommodation.

13. Cross-Linguistic Calibration

The architecture makes typological predictions. If nominal potential is a licensing property of syntactic role systems, then languages with robust nominalization strategies should show a larger inventory of NP_syn members. Conversely, languages that restrict clause-like materials from argument positions should display tighter NP_syn inventories and, correlatively, tighter distributions of NP_syn-level operators.

13.1 External negation and constituent operators

A hypothesis consistent with the two-level view is that some external or constituent negation strategies are best modeled as operators over nominal candidates, rather than as uniform clausal heads.

13.2 Calibration targets

The expanded typology chapter in this edition operationalizes this hypothesis with explicit diagnostics and language-by-language sketches.

14. Why Uniform NP/DP Implementations Overgenerate

14.1 NP is lexical, not syntactic

If NP is defined lexically as noun-headed phrase, it cannot simultaneously be the syntactic class of all nominal hosts. Treating NP as both forces ad hoc decisions about whether an infinitival or clause is really an NP.

14.2 DP does not predict NOT

DP theory predicts that determiners select NP_lex. If NOT were a determiner head in the DP system, it would be expected to select NP_lex; but constituent NOT selects complements that are not NP_lex.

14.3 Overgeneration risks

Once silent nominal structure is freely available, almost any constituent can be forced into DP. The NP_syn analysis retains explicit limits: only constituents licensed for nominal hosting are eligible for NP_syn-level NOT.

15. Diachrony: Negation and Nominalization

15.1 Negation cycles

Negation often cycles through reinforcement and reanalysis (Jespersen 1917; Horn 1989; Dahl 2010). The present framework highlights a diachronic hypothesis: constituent-negation uses arise naturally in focus and contrast structures, where operators target constituents that are independently hostable as nominal candidates.

15.2 Nominalization as expansion of NP_syn

Nominalization strategies can be unified as expansions of NP_syn: they increase the number of constituents that can be licensed as nominal candidates. This predicts a coupling between the growth of nominalization devices and the availability of NP_syn-level operator systems.

16. Computational and Psycholinguistic Implications

16.1 Parsing architecture

A two-level architecture suggests a division of labor for parsers: (i) a lexical component that builds head-driven local structures with selectional constraints, and (ii) a role-licensing component that assigns constituents to subject/predicate/object slots and computes nominal-potential coercions.

16.2 Predicting attachment and scope

Treating constituent NOT as Det_syn predicts that its attachment decisions depend on nominal hosting and information structure rather than on the clausal spine alone. An implementable heuristic is to prefer NOT attachment to the smallest constituent that is independently hostable as NP_syn in the local construction.

16.3 Processing

The model predicts increased processing cost for NP_syn uses that require stronger coercion (e.g., AP/PP in subject position), and it predicts that NOT will be more acceptable with those materials in discourse contexts that explicitly establish a contrast set.

17. Conclusion and Future Work

The two-level architecture and the NP split yield a compact explanation for the broad constituent distribution of English NOT: NOT is a syntactic determiner over NP_syn, not a lexical determiner over NP_lex and not exclusively a clausal head. The rule NP_syn -> NOT NP_syn captures the observed complement diversity, allows limited recursion, and prevents competition with articles.

The expanded chapters that follow provide an empirical construction atlas and a clearer boundary between NP_syn-level NOT and clausal negation. Future work should (i) operationalize nominal-potential diagnostics in corpus studies, (ii) test whether NOT-like operators in other languages align with NP_syn membership, and (iii) explore computational implementations of the two-level interface.

Limitations and anticipated objections. First, free-adjunction analyses may argue that not attaches broadly and that apparent constraints are pragmatic or prosodic. The present account differs by encoding an explicit selectional constraint at the syntactic level (NP_syn), yielding

falsifiable predictions about where constituent not is unavailable even under supportive discourse. Second, ellipsis and fragment diagnostics can obscure underlying structure (e.g., clausal sources for short answers). The monograph therefore treats ellipsis facts as diagnostics to be triangulated with hosting and coordination, not as decisive evidence on their own. Third, register and constructional markedness matter: some NP_syn hosts (especially AP/PP in nominal positions) are facilitated by specific information-structural frames (clefts, contrast topics) and may vary across speakers. The predictions are stated at the level of licensing availability, not uniform acceptability. Finally, polarity licensing is primarily a clause-level phenomenon; the theory predicts that NP_syn-level not will not automatically reproduce NegP-level NPI licensing effects without independent clausal negation. These points define the empirical tests that most directly discriminate the NP_syn/Det_syn proposal from spine-only (NegP) and attachment-only baselines.

18. Construction Atlas: A Glossed Example Set by Construction Type

This chapter provides a larger, consistently glossed example set organized by construction type. Unless otherwise indicated, the examples are constructed to illustrate structural contrasts and are intended to be evaluated by native-speaker judgment. The glossing conventions are given in Appendix D.

18.1 NP_lex complements

- (18) a. not the cat (but the dog)
 NEG DET cat
 'not the cat'
 b. not a doctor (but a nurse)
 NEG INDF doctor
 'not a doctor'
 c. not my neighbor (but my colleague)
 NEG POSS neighbor
 'not my neighbor'
 d. *the not cat
 DET NEG cat
 intended: 'the non-cat' (blocked: NOT is not Det_lex)

18.2 Infinitival NP_syn (InfP_syn)

- (19) a. Not to swim is dangerous.
 NEG INF swim COP dangerous
 'Not swimming is dangerous.'
 b. I decided not to leave early.
 1SG decide.PST NEG INF leave early
 'I decided not to leave early.'
 c. Not to speak is sometimes prudent.

NEG INF speak COP sometimes prudent
'Not speaking is sometimes prudent.'

The relevant point is structural: InfP_{syn} is licensed as NP_{syn} in subject/complement positions, and NOT targets that nominal candidate. Register and verb-class restrictions on infinitival complements (e.g., prefer, try, decide) remain lexical/selectional and do not undermine the NP_{syn} generalization.

18.3 Clausal NP_{syn} (Clause_{syn})

- (20) a. Not that he left bothered me.
NEG COMP 3SG leave.PST bother.PST 1SG.OBJ
'It was not the fact that he left that bothered me.'
b. Not because he was tired, but because he was bored, he left.
NEG because 3SG COP.PST tired but because 3SG COP.PST bored 3SG
leave.PST
'He left not because he was tired but because he was bored.'

These cases are contrastive and discourse-structured. They illustrate that some CP-like constituents can be hosted as discourse nominals (topics, contrastive pivots) and are therefore eligible for NP_{syn}-level NOT.

18.4 PP NP_{syn} in focus/topic frames

- (21) a. Not in the house is where I found it.
NEG in DEF house COP where 1SG find.PST 3SG.OBJ
'It was not in the house that I found it.'
b. Not on Tuesday, but on Wednesday, they met.
NEG on Tuesday but on Wednesday 3PL meet.PST
'They met not on Tuesday but on Wednesday.'

18.5 AP NP_{syn} via copular/cleft-like hosting

- (22) a. Not ready is how he seemed.
NEG ready COP how 3SG seem.PST
'He seemed not ready.'
b. Not happy, but relieved, was her expression.
NEG happy but relieved COP POSS expression
'Her expression was not happy but relieved.'

18.6 Participial/eventive NP_{syn} (PartP_{syn})

- (23) a. Not sitting in silence was the problem.
NEG sit.PROG in silence COP DEF problem
'The problem was (specifically) not sitting in silence.'
b. Not answering the email immediately was intentional.
NEG answer.GER DEF email immediately COP intentional
'The lack of immediate reply was intentional.'

18.7 Coordination and ellipsis

- (24) a. Not to swim and not to run were both impossible.
NEG INF swim and NEG INF run COP.PST both impossible
'Both refraining from swimming and refraining from running were impossible.'
- b. She chose not to call, and he did too. (ellipsis)
3SG.F choose.PST NEG INF call and 3SG.M do.PST too
'He also chose not to call.'

Coordination supports the NP_syn analysis because the coordinated units behave like nominal candidates for hosting (as subjects or complements), even when their internal categories differ.

18.8 Recursion and contrast sets

- (25) a. not (not the cat) but the dog
NEG (NEG DET cat) but DET dog
'not (not the cat) but the dog' (double contrast)
- b. Not not to leave - but to stay - was the plan.
NEG NEG INF leave but INF stay COP DEF plan
'The plan was to stay (contrary to a prior exclusion).'

Nested NOT is predicted structurally by NP_syn recursion. Its discourse felicity depends on whether multiple contrast sets are pragmatically accommodated.

19. Constituent NOT vs Clausal Negation: Auxiliaries, Scope, and Polarity

English not has at least two grammatically distinct regimes. Clausal negation targets propositional content and is tightly integrated with the auxiliary system (including do-support and contraction). Constituent NOT, in the sense analyzed here, is an NP_syn-level operator that packages a nominal candidate as excluded relative to a contrast set. This chapter clarifies diagnostics separating these regimes and shows how the NP_syn analysis predicts their divergent behavior.

19.1 Structural position and auxiliary support

Clausal negation in English is expressed in the clausal spine: not (or n't) appears in the extended verbal domain, typically in association with auxiliaries (Haegeman 1995). In finite clauses without an auxiliary, do-support surfaces (Klima 1964).

- (26) a. John did not leave.
John do.PST NEG leave
'John did not leave.'
- b. John didn't leave.
John do.PST-NEG leave
'John did not leave.'

By contrast, NP_syn-level NOT is not tied to do-support or to auxiliary placement because it does not target the clausal spine. Its distribution is determined by whether its complement is licensed as NP_syn in the relevant construction.

19.2 Scope diagnostics

Clausal negation normally takes scope over the proposition, interacting with quantifiers and modality. Constituent NOT takes scope over a constituent and supports corrective contrast (Horn 1989). The two can be disambiguated by paraphrase and by the availability of focus continuations (not X, but Y).

- (27) a. John did not invite a doctor. (NEG > $\exists$, typical)
b. John invited not a doctor, but a nurse. (contrastive constituent NEG)
c. Not a doctor called - a nurse did. (NP_syn-level NOT on subject)

19.3 Polarity licensing

A central property of clausal negation is polarity licensing: negative environments license NPIs such as any, ever, and at all (Ladusaw 1979; Giannakidou 1998). Constituent NOT does not automatically create a downward-entailing clausal environment and therefore does not uniformly license NPIs.

- (28) a. John didn't see anything. (NPI licensed under clausal negation)
b. *Not John saw anything. (degraded: constituent negation does not license NPI)
c. John saw not anything, but something. (requires special focus; not equivalent to (a))

The contrast is expected if NP_syn-level NOT is an operator over nominal candidates rather than a propositional negator. Polarity phenomena should therefore be modeled in the clausal system (NegP, agreement, concord) rather than by appealing to NP_syn-level NOT.

19.4 Interaction with negative determiners

Many languages realize a negative determiner (e.g., German kein, French aucun, English no) that directly composes with NP_lex and encodes quantificational negation. English no is not identical to constituent NOT: no is a determiner in the lexical determiner system, while NOT in this monograph is Det_syn, an operator in the syntactic determiner system.

- (29) a. no doctor called.
NEG.DET doctor call.PST
'No doctor called.'
b. not a doctor called (but a nurse).
NEG INDF doctor call.PST
'Not a doctor called (but a nurse did).'

19.5 Summary diagnostic table

Table 1 summarizes practical diagnostics for distinguishing the two regimes in English.

Diagnostic	Clausal negation (Neg_clausal)	Constituent NOT (Det_syn over NP_syn)
Auxiliary/do-support	Typically yes (do-support in finite clauses without auxiliary)	No (depends on NP_syn hosting, not auxiliaries)
Primary scope	Proposition (p)	Nominal candidate (x in e*)
NPI licensing	Robust (licenses any/ever/at all)	Not automatic; requires separate licensing
Corrective continuation	Possible but less diagnostic	Canonical: not X, but Y
Position	Clausal spine (around T/Aux)	Immediately before NP_syn complement

20. Typology: NP_syn Diagnostics Applied Language-by-Language

This chapter operationalizes NP_syn membership diagnostics across languages and uses them to formulate predictions about the distribution of constituent-negation operators. The goal is not exhaustive typology but a replicable protocol: identify which non-noun constituents can be hosted as arguments or discourse nominals, then test whether negation operators pattern with that host class.

20.1 A diagnostic protocol for NP_syn

For a candidate phrase type X (infinitival, nominalized clause, PP, AP, etc.), evaluate:

- D1. Subject hosting: can X occur as the subject of a copular/predicational clause?
- D2. Object hosting: can X occur as a selected complement where nominal arguments occur?
- D3. Discourse hosting: can X be topicalized/contrastively fronted as a discourse pivot?
- D4. Operator compatibility: can X combine with a constituent-negation/focus operator (NOT-like) while preserving hostability?
- D5. Coordination/parallels: can X coordinate with canonical nominals under appropriate readings?

On the present view, D1-D3 diagnose nominal potential directly; D4 tests whether a language has an NP_syn-level operator system; D5 checks re-identifiability under standard compositional operations.

20.2 English (baseline)

- (30) a. To swim is fun.
 INF swim COP fun
 'Swimming is fun.'

b. Not to swim is dangerous.
 NEG INF swim COP dangerous
 'Not swimming is dangerous.'
 c. Not in the house is where I found it.
 NEG in DEF house COP where 1SG find.PST 3SG.OBJ

English exhibits a relatively permissive NP_syn inventory (infinitivals, certain clauses, and some PPs/APs in specific frames) and an overt NP_syn-level operator NOT.

20.3 German

German distinguishes clausal negation nicht from the negative determiner kein. Constituent negation and corrective focus are also expressed with nicht, often in contrastive frames.

(31) a. Nicht der Arzt, sondern die Krankenschwester kam.
 NEG DEF doctor but DEF nurse come.PST
 'Not the doctor but the nurse came.'
 b. Kein Arzt kam.
 NEG.DET doctor come.PST
 'No doctor came.'
 c. Zu schwimmen macht Spass.
 to swim.INF make.PRS fun
 'To swim is fun.'

Prediction: if zu-infinitivals are freely hostable as subjects (D1) and if nicht can target such hosted constituents (D4), German should permit a constituent-negation pattern parallel to English not to V in appropriate constructions. Where German instead prefers morphological/structural nominalization, NP_syn membership is mediated by those devices.

20.4 French

French clausal negation is canonically expressed via ne ... pas, and French also has negative determiners (aucun) and constituent focus strategies that can approximate not X, but Y patterns.

(32) a. Jean n'est pas venu.
 Jean NEG COP.PRS NEG come.PTCP
 'Jean did not come.'
 b. Aucun docteur n'est venu.
 NEG.DET doctor NEG COP.PRS come.PTCP
 'No doctor came.'
 c. Ne pas nager est dangereux.
 NEG NEG swim.INF COP dangerous
 'Not swimming is dangerous.'

French makes heavy use of infinitival/nominalized configurations for argumental hosting; thus NP_syn membership for non-noun material is often explicit. The protocol predicts that constituent negation over those hosted units should be available where discourse structure supports correction.

20.5 Spanish

Spanish uses *no* for clausal negation and has negative determiners/pronouns (*ningun*, *nadie*). Infinitivals are readily hostable as subjects.

- (33) a. Juan no vino.
Juan NEG come.PST
'Juan did not come.'
b. No nadar es peligroso.
NEG swim-INF COP dangerous
'Not swimming is dangerous.'
c. Ningun medico vino.
NEG.DET doctor come.PST
'No doctor came.'

20.6 Japanese

Japanese has rich nominalization strategies (e.g., *koto/no*) that make clause-like material hostable as arguments. Negation is expressed morphologically on the predicate (*nai*), and negative nominalizations are straightforward.

- (34) a. oyog-u koto-wa tano-shii.
swim-NPST NMLZ-TOP fun-PRS
'Swimming is fun.'
b. oyog-anai koto-wa kiken da.
swim-NEG NMLZ-TOP dangerous COP
'Not swimming is dangerous.'

Because NP_syn membership is often made explicit by nominalizers, the diagnostic protocol predicts that Japanese constituent negation should appear primarily as negation internal to nominalized structures (NEG inside X) rather than as a separate preposed NOT-like determiner.

20.7 Korean

Korean similarly licenses argumental clause-like material via nominalizers (*kes*, *ci*), and negation is expressed periphrastically or morphologically. Negative nominalizations provide a direct test of the NP_syn expansion hypothesis.

- (35) a. suyengha-ci anh-nun kes-un wihyeomha-da.
swim-CI NEG-REL NMLZ-TOP dangerous-DECL
'Not swimming is dangerous.'

20.8 Turkish

Turkish expresses negation with a verbal suffix (*-mA*) and has productive non-finite nominalizations. Negative infinitivals provide clear NP_syn candidates.

- (36) a. yuz-me-mek tehlikeli-dir.
swim-NEG-INF dangerous-COP
'Not swimming is dangerous.'

20.9 Summary: typological expectations

Table 2 summarizes a coarse-grained application of the diagnostic protocol to a small language sample (illustrative).

Language	Argumental non-finites (D1/D2)	Argumental clauses via nominalizer	Dedicated NEG.DET	Preposed constituent-NOT operator	Notes
English	Yes (to-V, V-ing)	Limited (that-clauses; some frames)	Yes (no)	Yes (not)	Clear split: no = Det_lex, not = Det_syn
German	Yes (zu-V)	Yes (dass, nominalization options)	Yes (kein)	Yes (nicht)	nicht spans clausal + constituent; kein is determiner
French	Yes (INF)	Yes (de/que clauses; nominalization)	Yes (aucun)	Partial (pas in focus)	Negation is ne...pas; focus uses vary
Spanish	Yes (INF)	Yes (que-clauses)	Yes (ningun)	Partial (no in focus)	no is clausal; constituent readings via focus
Japanese	Yes (NMLZ koto/no)	Yes (koto/no)	No (as determiner)	No (preposed)	NEG is predicate morphology; NP_syn via nominalizers
Korean	Yes (kes/ci)	Yes (kes/ci)	No (as determiner)	No (preposed)	NEG periphrasis + nominalization
Turkish	Yes (INF/nominals)	Yes (various)	No (as determiner)	No (preposed)	NEG is suffix; NP_syn expansion is morphological

The table highlights a **hypothesis**: *languages with strong morphological nominalization may realize NP_syn expansion internally* (NEG inside nominalized clauses) rather than via a dedicated preposed NP_syn-level negator. Conversely, languages that allow preposed contrastive operators may show an English/German-style constituent negation pattern.

21. Related Work and Distinguishing Predictions

The sections below situate these claims in the negation literature and specify tests that discriminate them from attachment-only and spine-only baselines.

Core Claims (Chapter 21)

1. Claim 1 (NP split). NP is split into NP_lex (noun-headed phrase; lexical module) and NP_syn (nominal-potential host; syntactic role system) (see §§5.3–6.4).
2. Claim 2 (Constituent NOT as Det_syn). Constituent NOT is a syntactic determiner-like operator that selects NP_syn and returns NP_syn: $\text{NP_syn} \rightarrow \text{NOT NP_syn}$ (see §7.1; Ch.19).
3. Claim 3 (No Det_lex competition). Because NOT is not a Det_lex element, it cannot occupy the article slot inside NP_lex; it therefore yields not [Det_lex N] but excludes *[Det_lex not N] (see ex. (3); §8.2).
4. Claim 4 (Separation from clausal diagnostics). Clause-level properties (auxiliary/do-support, inversion, canonical NPI licensing) track clausal negation in the spine, and need not be reproduced by NP_syn-level NOT (see §§19.1–19.3; §21.6).
5. Claim 5 (Typological coupling). The availability of constituent-negation operators correlates with the size of NP_syn (nominalization resources and argument-hosting strategies) (see §20.1; Table 21.1).

21.1 Sentential negation in the clausal spine

A large literature treats clausal (sentential) negation as part of the functional architecture of the clause. In generative syntax, this is commonly implemented as a dedicated projection in the extended verbal domain (NegP or related heads), whose position and feature content interact with tense, auxiliaries, movement, and scope (Haegeman 1995; Zanuttini 1997; De Clercq 2013). In English, canonical clausal negation is realized in the auxiliary system (do-support in finite contexts without another auxiliary; n't as an affixal variant), and it participates in diagnostics that are sensitive to clause-level scope and functional structure (Huddleston & Pullum 2002).

Within this tradition, clausal negation is not merely a semantic operator; it is a structural ingredient with downstream consequences. Arguments from ellipsis and related licensing conditions, for instance, have been used to motivate a negation-related head in the clausal spine, because ellipsis licensing is often taken to require a head–complement configuration that a mere VP-adjunct would not provide (Potsdam 1997; Hankamer 2009). Likewise, classic tests distinguishing wide-scope negation from narrow-scope negation (e.g., inversion and certain polarity effects) typically presuppose a clause-level structural locus for sentential negation (Klima 1964; Huddleston & Pullum 2002).

21.2 Constituent negation and “two-lives” analyses of not

Alongside clausal negation, English displays a robust family of constituent-negation uses: contrastive fragments (Not John), corrective constructions (not X, but Y), and narrow negation of subclausal constituents (not in the garden, not to leave, not that he called). A common response in formal syntax is a “two-lives” view, on which not instantiates at least two structural realizations—one associated with clausal negation in the auxiliary domain and another as an adverbial or phrasal modifier that can attach to smaller constituents (Kim & Sag 1995; Kim 2000; Kim & Sag 2002; Kim 2001).

Two-lives approaches correctly capture the intuition that English not participates both in clause-level polarity systems and in narrower contrastive uses. However, they often leave a key question underconstrained: what predicts where constituent-level not can attach in a way that yields a stable constituent that is itself hostable as an argument or discourse topic? In practice, “phrasal modifier” analyses risk becoming permissive unless they are paired with an explicit licensing theory specifying which derived constituents are eligible for nominal hosting.

21.3 Maximal adjunction: not as a category-general phrasal adjunct

A stronger minimalist line proposes that English not (and n’t) can be treated as adjuncts throughout, eliminating a dedicated Neg head and deriving clausal vs. constituent differences from attachment height and scope alone (Bruening 2025). In its strongest form, not is taken to be able to attach to virtually any category, while the observed restrictions are explained through general constraints on scope, linearization, and the auxiliary system.

This monograph shares the motivation to avoid covert structure where it is not empirically forced. The concern is instead predictive: if not is freely attachable, then the theory must separately explain why only certain attachments yield constituents that behave like well-formed nominal candidates—units that can occupy subject/object/topic positions, coordinate with other nominal candidates, and be tracked in discourse. Absent an explicit nominal-host class, the analysis risks shifting from prediction to post hoc attachment choices, particularly in cases where the same surface sequence admits both clause-level and constituent-level parses with different diagnostics (Huddleston & Pullum 2002).

21.4 DP-internal negation and negative determiners

A further comparative baseline comes from negative determiners and negative indefinites in Germanic and beyond (e.g., German *kein*, Dutch *geen*, English *no*). These items are widely analyzed as determiner-like elements combining an indefinite component with negation, often displaying compositional behavior and, in some languages, split-scope readings (Jacobs 1980; Kratzer 1998; de Swart 2000). Their determiner status is visible in article competition, inflectional agreement, and their position in DP/NP structure.

This distinction is methodologically important for English. If not were simply a negative determiner of the *no/kein* type, it should behave as a member of the lexical determiner system:

competing for the article slot and participating in noun-phrase-internal morphology. English does not show this pattern. Instead, not freely precedes full noun phrases that already contain a lexical determiner (not the cat, not a problem), and it can also precede complements that are not noun-headed (not to leave, not that he called). The present account therefore distinguishes determiner-like behavior at two levels: (i) lexical determiners (Det_lex) inside NP_lex, and (ii) syntactic determiner operators (Det_syn) that package nominal candidates at the sentence-role interface.

21.5 Point of departure: nominal potential as a licensing property

The present proposal departs from both NegP-centered and pure-adjunction accounts in where it locates the key constraint on constituent-level not. The claim is not merely that English has clause-level and constituent-level negation—this is widely recognized. The claim is that the constituent-level distribution is best predicted by a separate licensing property: nominal potential. Nominal potential is defined as eligibility to occupy a nominal role in the syntactic module (subject/object/topic and related discourse-nominal hosts), independently of noun-headedness. This yields the NP split: NP_lex (noun-headed phrases computed in the lexical module) versus NP_syn (the functional class of phrases with nominal potential). Constituent not is then analyzed as a syntactic determiner operator Det_syn, mapping NP_syn to NP_syn.

21.6 Distinguishing predictions

Table 21.1 summarizes the distinguishing predictions of the NP_syn/Det_syn analysis relative to two common baselines. The tests are designed to be implementable as minimal pairs or controlled acceptability/interpretation tasks.

Table 21.1. Distinguishing predictions and diagnostic tests.

Prediction	NP_syn/Det_syn account	Adjunct-only baseline	NegP-only baseline	Diagnostic test
P1 Licensing-by-hosting	NOT selects NP_syn only; complement diversity tracks independent argument-hosting (nominal potential).	NOT adjoins broadly; constraints largely pragmatic/phonological; risks overgeneration.	NOT is spine negation; constituent uses require covert clausal sources or ellipsis.	Minimal pairs: allow NOT before X iff X is independently hostable as NP_syn (Ch.18; §6).
P2 No article-slot competition	NOT cannot occupy Det_lex position; predicts not [a/the N] but blocks *a/the [not N].	May predict DP-internal NOT unless blocked by extra linearization constraints.	Must treat *a not N either as independent constraint or as mismatch with clausal source.	Classic asymmetry tests (ex. (3)); extend to possessives/demonstratives.
P3 Clause diagnostics separation	Aux/do-support, inversion, clause-wide NPI licensing are properties of clausal negation, not NP_syn-level NOT.	Adjunct view may blur clause/phrase boundary and mispredict NPI licensing from subclausal NOT.	If all NOT is NegP, predicts stronger coupling of diagnostics to NOT presence.	NPI tests (any/ever) and auxiliary diagnostics in matched contexts (Ch.19).
P4 Closure and coordination	Det_syn maps NP_syn→NP_syn; NOT+X should coordinate with other	Coordination depends on attachment site; may allow category-	Requires deriving NOT+X as clausal;	Coordination/ellipsis paradigms (Ch.18 §18.7; §21.6).

	NP_syns and behave as a nominal candidate.	mismatched coordination too freely.	coordination predicted only via ellipsis or restructuring.	
P5 Attachment minimality	Prefer smallest available NP_syn host in ambiguous strings; predicts systematic attachment biases.	Adjunction predicts multiple attachments with weak grammatical preference.	NegP predicts attachment to clausal spine unless special constituent negation mechanism added.	Prosody-controlled ambiguity tasks; compare interpretation/acceptability under controlled contexts.
P6 Typological coupling	Richer nominalization/argument-hosting expands NP_syn; predicts wider availability of constituent-negation operators.	No necessary typological link predicted.	Negation tied to clausal spine; constituent operators expected to be derived, not correlated with nominalization.	Apply protocol in §20.1: NP_syn diagnostics → check constituent-negation operator distribution.

21.7 Summary and evaluation program

In summary, the novelty claim of the present analysis is not that not has multiple uses, but that constituent-level distribution is governed by a specific interface condition: nominal potential. This yields a concrete evaluation program: (i) operationalize NP_syn membership using independent diagnostics (Chapter 20); (ii) test whether constituent-level NOT distribution tracks NP_syn membership more tightly than attachment-based alternatives; and (iii) measure how clausal diagnostics (inversion, polarity licensing, ellipsis) partition clausal NOT from NP_syn-level NOT. If these correlations hold, the two-level NP split is not just a terminological re-labeling but a constraint-bearing refinement of the grammar of negation.

22. Computational Applications

This chapter develops applied computational consequences of the NP_syn/Det_syn analysis of NOT. Chapter 16 framed general processing implications; here the goal is engineering-facing: how the two-level grammar can be operationalized for parsing, scope/focus resolution, information extraction, and evaluation, with explicit points of contact to established negation resources and shared tasks (Chapman et al. 2001; Szarvas et al. 2008; Morante and Blanco 2012; Morante and Daelemans 2009; Peng et al. 2018).

22.1 From diagnostics to computable annotations

Most NLP pipelines treat negation as a cue-and-scope phenomenon. This is effective for many downstream tasks, but it typically presupposes that negation is clausal. The present monograph adds a second target: NP_syn-level negation, where NOT acts as Det_syn selecting a nominal-potential host. Computationally, this yields a concrete annotation proposal: for each NOT token, (i) classify it as clausal-negation vs NP_syn-negation (or allow a third class 'metalinguistic' if desired), and (ii) for NP_syn-negation, identify the minimal NP_syn host and its realized complement. The key claim is that host identification is not equivalent to phrase category labeling; it depends on nominal potential diagnostics (Chapter 6) that can be approximated with syntactic and distributional features.

22.2 Two-level parsing: lexical headhood vs role-licensing

A direct implementation strategy is a two-pass parser. Pass 1 derives a standard phrase structure or dependency representation sufficient to identify NP_lex, clause boundaries, auxiliaries, and basic attachment. Pass 2 computes NP_syn membership as a derived feature of constituents based on (a) argument-position availability, (b) coordination closure, (c) operator compatibility, and (d) identity and re-identifiability heuristics (Chapter 6). In this view, NP_syn is not a primitive tag but a computed property of spans or nodes; NOT-as-Det_syn is then predicted when NOT immediately composes with an NP_syn node, returning NP_syn (Chapter 7).

A practical approximation can be built on top of Universal Dependencies (UD) parses by learning a classifier over NOT occurrences. Features include: governor category (aux vs content predicate vs nominal head), dependency configuration (neg attached to AUX/VERB vs attached to ADJ/ADV/ADP/clausal complement), local determiners (presence of Det_lex inside NP_lex), coordination patterns, and the availability of a nominalized reading in nearby argument positions. This complements rule-based systems such as NegEx/ConText (Chapman et al. 2001; Harkema et al. 2009) by explicitly targeting constituent-NOT and by distinguishing host licensing from mere surface proximity.

22.3 Scope and focus resolution with NP_syn hosts

Shared tasks in computational semantics have operationalized 'scope' and 'focus' of negation, providing evaluation data and metrics (Morante and Blanco 2012). Biomedical and clinical corpora extend this with scope annotations and uncertainty (Szarvas et al. 2008; Vincze et al. 2008), motivating systems that combine syntactic constraints and learned models (Morante and Daelemans 2009). The NP_syn/Det_syn analysis suggests a refinement: for NP_syn-level NOT, the primary target is not a clausal scope but a host span. Thus, a system should output Host(NOT)=NP_syn-span and optionally Focus(NOT)=subspan within the host if a focus model is desired.

This refinement is especially relevant for clinical text. In radiology, 'no evidence of X' is often handled well by cue-and-scope methods, but constituent negation arises in nominalized event descriptions and adjective/PP frames (e.g., 'not applicable', 'not in the lung fields') that function argumentally. Dependency-pattern approaches such as NegBio already exploit UD structure to delineate scope (Peng et al. 2018); the present proposal adds a principled host-typing layer, predicting when NOT should be treated as selecting a nominal candidate rather than introducing clausal negation.

22.4 Downstream tasks: extraction, QA, and translation

Information extraction. Systems that populate knowledge bases from text require correct polarity for entities and events. If NOT selects NP_syn, the negated object is often best modeled as a nominalized description (a 'thing that is not X') rather than as a negated proposition. This distinction matters for relation extraction and entity linking in domains where descriptions are treated as referents.

Question answering. Many QA errors stem from treating negated constituents as if they were positive mentions (or vice versa). A host-based representation supports explicit reasoning: NOT modifies an NP_syn candidate, so the query match should exclude the modified predicate within that nominal description while preserving the referential shell.

Machine translation. Languages differ in where negation is realized (sentential particles, affixes, negative concord). If constituent negation is packaged as NP_syn-level Det_syn in English, the translation system can map it to the target language's nominal negation strategy (e.g., negative determiners or morphological negation) rather than forcing a clausal strategy. This aligns with typological expectations that nominalization resources correlate with constituent-negation operators (Chapter 20; de Swart 2010).

22.5 Evaluation: a test suite for NP_syn-level NOT

Beyond corpus F1, the monograph motivates a grammar-driven evaluation suite. Minimal-pair tests can be generated from the Distinguishing Predictions table (Table 21.1) by constructing pairs that differ only in the availability of NP_syn hosting (e.g., not + NP_lex vs *Det_lex + not

+ N; not + AP/PP in argument position vs the same AP/PP without argumental packaging). Such suites complement shared-task corpora by directly testing the theory's selectional claim.

Recommended metrics include: (i) accuracy of NOT-type classification (clausal vs NP_syn), (ii) exact-match host-span accuracy for NP_syn cases, (iii) downstream polarity impact (e.g., effect on extracted assertions), and (iv) robustness across registers (formal prose, dialogue, clinical notes).

22.6 Implementation sketch

A practical baseline is a hybrid system: UD parsing + deterministic heuristics for high-precision NP_syn host identification + a learned classifier to resolve ambiguous cases. Training data can be created by annotating a modest corpus with the NP_syn diagnostics in Chapter 6. Neural sequence models can be used as back-ends, but the key contribution is the representational target: host-typed negation as NP_syn selection, not only scope in the clause. This target is compatible with existing negation corpora and can be layered on them.

23. Neuroscience and Neurocognitive Implications

This chapter connects the two-level analysis of NOT to experimental and neurocognitive work on negation. A central theme in the neuroscience of negation is whether comprehension proceeds via a delayed or two-step mechanism and how pragmatic support modulates processing cost (Fischler et al. 1983; Kaup et al. 2007; Nieuwland and Kuperberg 2008). The NP_syn/Det_syn proposal adds a structural factor: some NOT occurrences are clause-level operators, while others are NP_syn-level packaging operators, and these may recruit partially distinct processing routines.

23.1 Behavioral signatures and processing architectures

Classic behavioral results often report longer verification times and increased error rates for negated sentences, motivating two-step accounts in which comprehenders first represent an affirmative core and only later integrate negation. However, other lines of work show that contextual support can reduce or eliminate such delays, suggesting more dynamic integration (Kaup et al. 2007; Tian et al. 2010; Nieuwland and Kuperberg 2008). The present monograph reframes a portion of this variability as structural: constituent-NOT (NP_syn-level) is predicted to behave like a packaging operator over a nominal candidate, whereas clausal NOT targets propositional content. If so, some 'negation costs' may reflect the extra work of building or selecting an NP_syn host, not only truth-value reversal.

23.2 ERP evidence: N400 modulation and pragmatic licensing

ERP work has been central in assessing the time course of negation integration. Fischler et al. (1983) reported ERP differences during sentence verification with negative forms, and later work demonstrated that pragmatic licensing crucially shapes N400 effects for negated statements (Nieuwland and Kuperberg 2008). In particular, pragmatically supported negation can show

N400 patterns more similar to affirmative sentences, whereas unsupported negation can attenuate truth-value sensitivity at the critical word. More broadly, the N400 literature emphasizes sensitivity to expectancy and semantic access during comprehension (Kutas and Federmeier 2011).

Predictions from NP_syn/Det_syn. If constituent-NOT is a determiner-like operator selecting NP_syn, it should be possible to observe ERP differences between (i) clausal negation environments (where polarity licensing and truth-value computation are engaged) and (ii) NP_syn-level negation environments (where the main operation is packaging a nominal description). The theory predicts that NP_syn-level NOT will not necessarily induce the same clause-wide polarity effects and may instead pattern with nominalization and referential packaging demands (Chapter 6; Chapter 19).

23.3 fMRI and inhibition: action systems and frontal control

Functional neuroimaging has linked negation to control and inhibition mechanisms. Tettamanti et al. (2008) reported modulation of action-related representations during comprehension of negative action sentences, consistent with the view that negation can down-regulate the simulated state of affairs. Other fMRI work contrasts negative and affirmative sentences and reports differences across temporal and frontal regions (Christensen 2009). These results are compatible with an architecture in which clause-level negation engages control mechanisms for truth-conditional revision, while NP_syn-level NOT engages selection and packaging mechanisms that resemble nominalization and referential operations.

23.4 Experimental designs to dissociate clausal vs NP_syn-level NOT

The monograph motivates a family of experiments in which structure is manipulated while surface form is held constant. Representative designs include:

- EEG/ERP minimal pairs: compare NOT with clearly clausal diagnostics (auxiliary support, inversion environments) to NOT with NP_syn hosts (argument-position AP/PP/infinitival packaging). Measure N400/P600 timing and truth-value sensitivity.
- Eye-tracking: test whether fixation dynamics differ when NOT selects NP_syn vs when NOT introduces clausal negation, controlling for length and lexical frequency.
- fMRI: contrast action-related clausal negation (as in Tettamanti et al. 2008) with NP_syn-level negation in nominalized event descriptions, predicting reduced engagement of action simulation but increased engagement of selection/packaging networks.
- Polarity manipulation: introduce NPIs and polarity-sensitive items to verify the prediction that NPI licensing is tied to clausal negation rather than to NP_syn-level NOT (Giannakidou 1998; Penka and Zeijlstra 2010).

23.5 Development, acquisition, and clinical relevance

Developmental work shows that children's comprehension of negation is strongly context-dependent (Nordmeyer and Frank 2014). The NP_syn/Det_syn account predicts that early

competence with truth-functional clause negation can diverge from competence with constituent negation, because constituent negation presupposes nominal-potential packaging and argument hosting. Clinically, the same distinction may matter when assessing negation comprehension in populations with executive-control limitations: clause-level negation may be disproportionately impaired relative to constituent packaging, or vice versa, depending on which control and selection mechanisms are affected.

23.6 Summary and research agenda

Neurocognitive work on negation often treats NOT as a single operator whose effects vary with context. The two-level account introduced here predicts a principled structural split: clausal NOT participates in polarity licensing and propositional truth-value reversal, whereas NP_{syn}-level NOT packages a nominal candidate and need not trigger clausal polarity behavior. The most direct empirical tests therefore compare matched stimuli that differ only in whether a nominal-potential host is available. This agenda links theoretical diagnostics (Chapter 6; Table 21.1) to measurable processing signatures across methods.

Appendix A. Notation

NP_{lex} : noun-headed phrase (lexical module)
NP_{syn} : phrase with nominal potential (syntactic module)
Det_{lex} : lexical determiners (articles, possessives, negative determiner no)
Det_{syn} : syntactic determiners (operators over NP_{syn}), with NOT as the central case
e* : generalized argument type associated with nominal potential
NEG.DET : negative determiner (cross-linguistic)

Appendix B. Master BNF Specification

For convenience, the master grammar is reproduced here as a single block.

```
S ::= Subject Predicate (Object)
Subject ::= NP_syn
Object  ::= NP_syn
Predicate ::= VP_syn

Det_syn ::= NOT

NP_syn ::= NP_lex
        | InfP_syn
        | Clause_syn
        | PP_lex
        | AP_lex
        | PartP_syn
        | Det_syn NP_syn
```

```

NP_lex ::= (Det_lex) N (Mod_lex)*
Det_lex ::= a | the | this | that | these | those | my | your | his | her
| its | our | their | Ø | no
Mod_lex ::= AP_lex | PP_lex
PP_lex ::= P NP_lex
AP_lex ::= A (PP_lex)?

InfP_syn ::= to V (NP_syn)? (PP_lex)*
PartP_syn ::= V-ing (NP_syn)? (PP_lex)*
Clause_syn ::= that S | because S | when S | if S

VP_syn ::= V
| V NP_syn
| V PP_lex
| V NP_syn PP_lex
| V Clause_syn

```

Appendix C. Extended Derivations

C.1 Nested NOT

NP_syn -> NOT NP_syn -> NOT (NOT NP_syn) -> not (not X)

Interpretation is compositionally available; pragmatic felicity depends on contrast-set accommodation.

C.2 Clause + infinitive hybrid

not [that he tried [to swim]]

Clause_syn hosts an embedded infinitival; NOT applies to the clausal nominal as NP_syn.

Appendix D. Glossing Abbreviations

1SG / 3SG / 3PL	: person/number
AUX	: auxiliary
CI	: Korean connective/nominalization marker (traditional label)
COMP	: complementizer
COP	: copula
DECL	: declarative
DEF	: definite
DET	: determiner
INDF	: indefinite article
INF	: infinitive marker
NEG	: negation (general)
NEG.DET	: negative determiner

NMLZ	: nominalizer
NPST	: non-past
OBJ	: object
POSS	: possessive
PROG	: progressive
PST	: past
REL	: relativizer
TOP	: topic marker

References

- Abney, Steven. 1987. The English noun phrase in its sentential aspect. PhD dissertation, MIT.
- Baker, Mark. 2003. Lexical categories: Verbs, nouns, and adjectives. Cambridge University Press.
- Bartoli, Elisabetta, et al. 2013. The disembodiment effect of negation: Negating action-related sentences attenuates corticospinal excitability. *Journal of Neurophysiology* 109(5): 1394–1401.
- Bruening, Benjamin. 2025. A simpler analysis of English negation (and the Bulgarian definite marker). Manuscript, University of Delaware.
- Chapman, Wendy W., Will W. Bridewell, Paul Hanbury, Gregory F. Cooper, and Bruce G. Buchanan. 2001. A simple algorithm for identifying negated findings and diseases in discharge summaries. *Journal of Biomedical Informatics* 34(5): 301–310.
- Chomsky, Noam. 1970. Remarks on nominalization. In Roderick A. Jacobs & Peter S. Rosenbaum (eds.), *Readings in English transformational grammar*, 184–221. Ginn.
- Christensen, Ken Ramshøj. 2009. Negative and affirmative sentences increase activation in different areas in the brain. *Journal of Neurolinguistics* 22(1): 1–17.
- Croft, William. 2001. *Radical Construction Grammar: Syntactic theory in typological perspective*. Oxford University Press.
- Dahl, Östen. 2010. Typology of negation. In Laurence R. Horn (ed.), *The expression of negation*, 9–38. De Gruyter.
- De Clercq, Karen. 2013. A unified syntax of negation. PhD dissertation, Ghent University.
- de Swart, Henriette. 2000. Scope ambiguities with negative quantifiers. In Klaus von Heusinger & Urs Egli (eds.), *Reference and anaphoric relations*, 109–132. Springer.
- de Swart, Henriette. 2010. *Expression and interpretation of negation: An optimality-theoretic typology*. Springer.
- Fischler, Ira, Paul A. Bloom, Diane G. Childers, Samuel E. Roucos, and Nancy W. Perry. 1983. Brain potentials related to stages of sentence verification. *Psychophysiology* 20(4): 400–409.
- Giannakidou, Anastasia. 1998. Polarity sensitivity as (non)veridical dependency. John Benjamins.

- Haegeman, Liliane. 1995. *The syntax of negation*. Cambridge University Press.
- Hankamer, Jorge. 2009. *Auxiliaries and negation in English*. Manuscript, University of California, Santa Cruz.
- Harkema, Henk, John N. Dowling, Tyler Thornblade, and Wendy W. Chapman. 2009. ConText: An algorithm for determining negation, experiencer, and temporal status from clinical reports. *Journal of Biomedical Informatics* 42(5): 839–851.
- Horn, Laurence R. 1985. Metalinguistic negation and pragmatic ambiguity. *Language* 61(1): 121–174.
- Horn, Laurence R. 1989. *A natural history of negation*. University of Chicago Press.
- Huddleston, Rodney & Geoffrey K. Pullum. 2002. *The Cambridge Grammar of the English Language*. Cambridge University Press.
- Jacobs, Joachim. 1980. Lexical decomposition in Montague grammar. *Theoretical Linguistics* 7: 121–136.
- Jespersen, Otto. 1917. *Negation in English and other languages*. København: A. F. Høst.
- Kaup, Barbara, Jana Lüdtkke, and Rolf A. Zwaan. 2006. Processing negated sentences with contradictory predicates: Is a door that is not open mentally closed? *Journal of Pragmatics* 38(7): 1033–1050.
- Kaup, Barbara, Rolf A. Zwaan, and Jana Lüdtkke. 2007. Experiential simulations of negated text information. *Quarterly Journal of Experimental Psychology* 60(7): 976–990.
- Kim, Jong-Bok. 2000. *The grammar of negation: A constraint-based approach*. CSLI Publications.
- Kim, Jong-Bok. 2001. Negation, VP ellipsis, and VP fronting in English: A Construction-HPSG analysis. Paper, Kyung Hee University. (Available via ACL Anthology.)
- Kim, Jong-Bok & Ivan A. Sag. 1995. The parametric variation of English and French negation. In José Camacho, Lina Choueiri & Maki Watanabe (eds.), *Proceedings of the Fourteenth West Coast Conference on Formal Linguistics (WCCFL 14)*, 303–317. CSLI Publications.
- Kim, Jong-Bok & Ivan A. Sag. 2002. Negation without head-movement. *Natural Language & Linguistic Theory* 20(2): 339–412.
- Klima, Edward. 1964. Negation in English. In Jerry A. Fodor & Jerrold J. Katz (eds.), *The structure of language*, 246–323. Prentice-Hall.
- Kratzer, Angelika. 1998. Scope or pseudoscope? Are there wide-scope indefinites? In Susan Rothstein (ed.), *Events and grammar*, 163–196. Kluwer.
- Kutas, Marta, and Kara D. Federmeier. 2011. Thirty years and counting: Finding meaning in the N400 component of the event-related brain potential (ERP). *Annual Review of Psychology* 62: 621–647.
- Ladusaw, William. 1979. *Polarity sensitivity as inherent scope relations*. PhD dissertation, University of Texas at Austin.

- Morante, Roser, and Eduardo Blanco. 2012. *SEM 2012 Shared Task: Resolving the Scope and Focus of Negation. In *SEM 2012: The First Joint Conference on Lexical and Computational Semantics, 265–274. Association for Computational Linguistics.
- Morante, Roser, and Walter Daelemans. 2009. A metalearning approach to processing the scope of negation. In Proceedings of the Thirteenth Conference on Computational Natural Language Learning (CoNLL 2009), 21–29. Association for Computational Linguistics.
- Nieuwland, Mante S., and Gina R. Kuperberg. 2008. When the truth is not too hard to handle: An event-related potential study on the pragmatics of negation. *Psychological Science* 19(12): 1213–1218.
- Nordmeyer, Ann E., and Michael C. Frank. 2014. The role of context in young children's comprehension of negation. *Journal of Experimental Child Psychology* 118: 25–39.
- Payne, John R. 1985. Negation. In Timothy Shopen (ed.), *Language typology and syntactic description*, Vol. 1: Clause structure, 197–242. Cambridge University Press.
- Peng, Yifan, Xiaosong Wang, Le Lu, Mohammadhadi Bagheri, Ronald M. Summers, and Zhiyong Lu. 2018. NegBio: A high-performance tool for negation and uncertainty detection in radiology reports. *AMIA Joint Summits on Translational Science Proceedings* 2018: 188–196.
- Penka, Doris & Hedde Zeijlstra. 2010. Negation and polarity: An introduction. *Natural Language & Linguistic Theory* 28(4): 771–786.
- Potsdam, Eric. 1997. NegP and subjunctive complements in English. *Linguistic Inquiry* 28(4): 533–541.
- Szarvas, György, Veronika Vincze, Richárd Farkas, and János Csirik. 2008. The BioScope corpus: Annotation for negation, uncertainty and their scope in biomedical texts. In *Proceedings of the Workshop on Current Trends in Biomedical Natural Language Processing*, 38–45. Association for Computational Linguistics.
- Tettamanti, Marco, et al. 2008. Negation in the brain: Modulating action representations. *NeuroImage* 43(2): 358–367.
- Tian, Ye, Barbara Kaup, and Rolf A. Zwaan. 2010. Why we simulate negated information: A dynamic account. *Quarterly Journal of Experimental Psychology* 63(12): 2305–2312.
- van der Wouden, Ton. 1997. *Negative contexts: Collocation, polarity and multiple negation*. Routledge.
- Vincze, Veronika, György Szarvas, Richárd Farkas, György Móra, and János Csirik. 2008. The BioScope corpus: Biomedical texts annotated for uncertainty, negation and their scopes. *BMC Bioinformatics* 9(Suppl 11): S9.
- Zanuttini, Raffaella. 1997. *Negation and clausal structure: A comparative study of Romance languages*. Oxford University Press.
- Zeijlstra, Hedde. 2004. *Sentential negation and negative concord*. PhD dissertation, University of Amsterdam.

